THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 1: **Exploration: Entering the World of Chemistry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Ignoring air resistance in a model of a falling ball is acceptable for a stone. Reason (R): For a stone, the effect of air resistance is very small compared with its weight. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. The mass of the Earth is about 6 x 10^24 kg and that of the Sun about 2 x 10^30 kg. The Sun is about ___ times heavier. 1

(a) 3 x 10^6 **(b)** 3 x 10^4 **(c)** 3 x 10^54 **(d)** 3 x 10^5

3. A scientist says 'The model fits in 99 of 100 trials, so the theory is proven'. A better statement is: 1

(a) the theory is false
(b) the theory is strongly supported but remains open to tests
(c) the theory is proven forever
(d) the data are wrong

4. Assertion (A): A scientific theory can never be changed. Reason (R): A theory is supported by a large amount of evidence. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A metre scale has a worn end and reads 0.3 cm at the start. A measured length reads 5.7 cm. The corrected length is: 1

(a) 5.4 cm **(b)** 0.3 cm **(c)** 6.0 cm **(d)** 5.7 cm

6. Which situation requires an exact value rather than an estimate? 1

(a) population of a country **(b)** number of stars in a galaxy
(c) height of a mountain in rough terms **(d)** dosage of a medicine for a child

7. Assertion (A): Estimation is useful in science. Reason (R): It gives a quick check on whether an answer is reasonable. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A stopwatch reads 20.4 s for 10 swings and each reading has an error of 0.2 s. The percentage error is about: 1

(a) 0.1 per cent **(b)** 1 per cent **(c)** 5 per cent **(d)** 10 per cent

9. **Assertion (A):** Repeating a measurement several times and taking the mean gives a more reliable value than a single reading. 1
Reason (R): Random errors in individual readings tend to cancel out when several readings are averaged.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The least count of a metre scale marked in millimetres is 1 mm. 1
Reason (R): The metre is the SI unit of length.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The sides of a rectangle are measured as 12.0 cm and 8.0 cm with an uncertainty of 0.1 cm each. Find the area and the percentage uncertainty. 2

12. Why does a log-log or straight-line graph help in checking a mathematical relationship? 2

13. Convert 1 light year into kilometres, given speed 3 x 10^5 km/s and 1 year = 3.15 x 10^7 s. 2

14. A student's data for g from five trials are 9.7, 9.9, 9.8, 9.6, 9.8 m/s^2. Find the mean and the range. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student measures a pencil three times as 14.2 cm, 14.3 cm and 14.1 cm. Find the mean length and the greatest deviation from it. 3

16. Explain why a scientist should report measurement uncertainties along with results with an example. 3

17. Explain how a graph of time period against length for a pendulum is expected to look. 3

OR

The time for 30 oscillations of a pendulum is measured three times: 54.2 s, 54.6 s and 54.4 s. Find the mean time period and the percentage spread. 3

18. Estimate the mass of air in a classroom of size 8 m x 6 m x 3 m (density of air 1.2 kg per cubic metre). 3

OR

Explain how science and ethics are linked using the example of testing medicines. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Kavya measures the length of a pencil with a ruler marked in millimetres. She takes five readings: 14.2 cm, 14.4 cm, 14.3 cm, 14.1 cm and 14.5 cm. Her teacher says that no single reading may be exactly right, so the readings should be combined. 5
(i) What is the least count of her ruler in cm?
(ii) Find the mean length of the pencil.
(iii) Express the mean length in metres.
(iv) Why is the mean more reliable than the first reading, and what is the largest deviation of any reading from the mean?

20. Discuss the strengths and limitations of the scientific method with five points and two examples of discoveries involving chance and persistence. 5

OR

The scientific community accepted heliocentrism after long debate. Describe how evidence, models and communication played a role. 5

END OF QUESTION PAPER — Total: 40 Marks